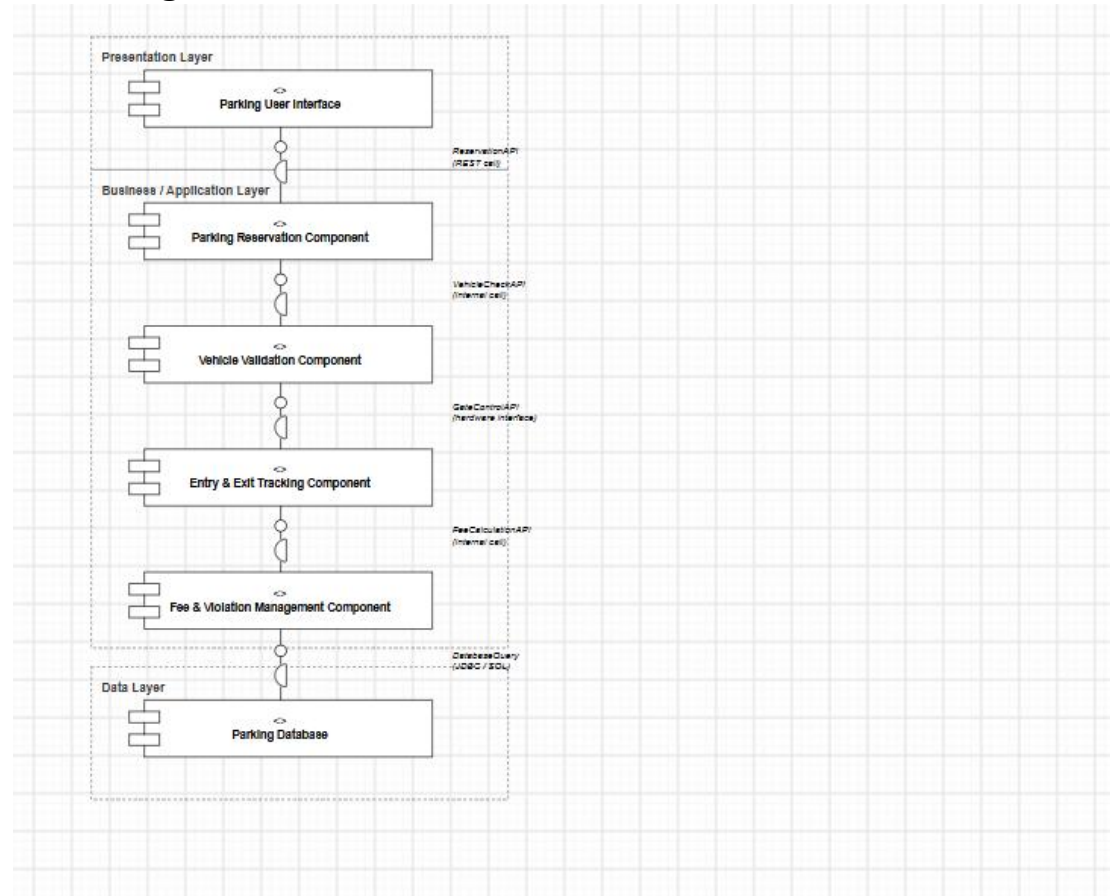


Lab 3: Component Modelling & Architectural Pattern Selection

UML diagram



1. Architecture Selection

Selected Architecture: Layered Architecture

We chose **Layered Architecture** for the Smart Parking System. The system is divided into three layers: **Presentation Layer**, **Business/Application Layer**, and **Data Layer**. The Presentation Layer contains the Parking User Interface. The Business/Application Layer contains the Parking Reservation, Vehicle Validation, Entry & Exit Tracking, and Fee & Violation Management components. The Data Layer contains the Parking Database.

2. Reasons for Selecting Layered Architecture

Reason 1 – Clear separation of parking functions:

The Smart Parking System performs several related functions such as parking reservation, vehicle validation, entry and exit tracking, fee calculation, and violation management. Layered Architecture separates these responsibilities into appropriate components, making the system easier to understand, develop, and maintain.

Reason 2 – Easy management of system operations:

The system receives requests from Vehicle Owners and Parking Wardens through the user interface and processes them through the business layer before accessing the database. This separation allows changes to one part of the system, such as the user interface or parking fee logic, to be made without directly changing the other layers.

3. Security Advantage

Layered Architecture provides a security advantage by keeping the **Parking Database separated from the user interface**. Users interact with the system through the Presentation Layer, while database access is handled through the Business/Application Layer. This reduces direct access to sensitive parking, vehicle, reservation, and violation information.

4. Performance Benefit

The layered structure allows the **Business/Application Layer to process requests before accessing the database**, reducing unnecessary direct database interactions. Vehicle validation, entry/exit tracking, fee calculation, and violation processing can therefore be handled by their respective components, helping the system respond efficiently to parking operations.